National University of Computer & Emerging Sciences
Karachi Campus

“E SHAADI”

Project Report
SOFTWARE DESIGN AND ANALYSIS
Section: J

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1. Executive Summary

E-Shaadi is a ground-breaking mobile matrimonial platform developed by **Group #03** from **FAST NUCES, Karachi**. The project aims to revolutionize the traditional "Rishta" (matchmaking) culture by transitioning it into a secure, digital environment. Unlike generic dating apps, E-Shaadi is specifically engineered to respect cultural nuances while leveraging modern technology to accelerate the search for a life partner.

The platform addresses critical issues in the current market—specifically lack of trust, limited reach, and inefficient manual searching—by implementing robust identity verification (CNIC/Passport) and AI-driven compatibility algorithms. With a scalable architecture built on Node.js and Flutter/React Native, and a monetization model based on premium subscriptions, E-Shaadi creates a safe, transparent ecosystem for brides, grooms, and their guardians.

2. Project Overview

2.1 Introduction and Purpose

The search for a compatible life partner is often a fragmented, time-consuming, and geographically limited process. **E-Shaadi** serves as a modern matchmaker, centralizing biodata, preferences, verification, and communication into a single intuitive mobile application.

Purpose:
The primary purpose of E-Shaadi is to digitize the matrimonial journey without losing the element of "trust" that is central to Eastern culture. It aims to:

- **Centralize Data:** Combine profile creation, preference setting, and matchmaking in one place.
- **Ensure Authenticity:** Prioritize user safety through rigorous document verification.
- **Modernize Tradition:** Bridge the gap between traditional family involvement and modern digital convenience by offering features like "Guardian Dashboards."

2.2 User Demographics & Problem Solving

To better understand the ecosystem E-Shaadi aims to build, the table below illustrates the target distribution of user profiles, emphasizing the goal of converting standard users into trusted, verified members.

Target User Profile Distribution

User Segment	Percentage Target	Description
Standard Users	45%	Unverified users browsing public content.
Verified Profiles	40%	Trusted users with Government ID verification (Badge awarded).

Premium Subscribers	15%	Paid users with access to unlimited swipes and advanced filters.

E-Shaadi identifies and bridges several gaps present in traditional matchmaking and existing digital alternatives:

Current Challenge / Problem	E-Shaadi Solution
Manual Fatigue ("Ghee Khatam") Traditional searches involve endless manual circulation of biodata, often leading to burnout without results.	**Automated Search Engine** Provides instant access to thousands of profiles with advanced filters, eliminating manual legwork.
Limited Network Reach Reliance on immediate family and "Rishta Aunties" limits options to specific social circles or neighborhoods.	**Expanded Digital Network** Connects users across cities and communities, vastly expanding the pool of potential partners beyond local networks.
Lack of Trust & Fake Profiles Digital alternatives are often plagued by scammers, fake profiles, and misleading information.	**Government ID Verification** Mandatory verification using CNIC, Passport, or Driver's License ensures profiles are authentic and trusted.
Privacy & Security Risks Photos and personal details often circulate uncontrollably on WhatsApp groups.	**Visibility Controls** Users control who sees their photos and data. Privacy settings allow visibility only to verified matches or specific groups.
Shallow Compatibility Decisions are often made on surface-level criteria (height, income) without understanding personality.	**AI Compatibility Scoring** Machine Learning algorithms analyze preferences, communication styles, and values to provide a deeper "Compatibility
	Score."

2.3 Team Roles and Responsibilities

The project is executed by a specialized team of five members, ensuring coverage across development, analysis, and operations.

Name	Role	Core Responsibilities
Azaan Hussain (Leader)	Backend Engineer	Focus on API reliability, data security, and system logic.
Haseeb Asif	AI Engineer	Focus on Smart Matching Algorithms and Model Training (TensorFlow/Scikit-learn).
Muzammil Siddiqui	Frontend Developer	Focus on Intuitive UI/UX, Mobile Screens, and Cross-Platform Design.
Rayyan Ali	DevOps Engineer	Focus on Automated Deployment (CI/CD), Cloud Infrastructure, and Dockerization.
Ahmed Ali	System Analyst	Focus on Requirement Elicitation, System Architecture, and Documentation.

2.4 User Experience (UX) & Design Philosophy

E-Shaadi is designed with a "User-First" approach, ensuring that users of all technical proficiency levels—from tech-savvy youth to traditional parents—can navigate the platform with ease.

User-Friendly Feature	Design Implementation	User Benefit
Simplified Onboarding	**Wizard-Style**	Reduces form fatigue and

	Registration: Profile creation is broken down into small, manageable steps (e.g., "Step 1 of 4").	ensures users complete their profiles without feeling overwhelmed by data entry.
Intuitive Navigation	**Bottom Navigation Bar:** Key sections (Home, Matches, Chat, Profile) are always one tap away.	Users can switch between browsing matches and checking messages instantly without getting lost in menus.
Visual Status Indicators	**Clear Badging System:** Uses distinct icons for "Verified," "Premium," and request statuses ("Pending," "Accepted").	Users can instantly assess the credibility of a profile or the status of a proposal at a glance.
Centralized Notifications	**Unified Alert Center:** All activities (New Matches, Interests, Messages) are aggregated in a single Notification tab.	Prevents users from missing critical updates and removes the need to constantly check different screens.
Guardian-Specific View	**Dedicated Dashboard:** A specialized interface optimized for parents managing multiple dependent profiles.	Simplifies the complexity for older generations by focusing only on relevant tasks like filtering and approving matches.

3. Technical Architecture

3.1 Technical Stack and Tools

The platform is built on a scalable, high-performance technology stack designed to handle secure data and real-time communication.

- **Frontend (Mobile):** React Native or Flutter for efficient cross-platform development (iOS & Android). Native modules (Kotlin/Swift) are used where specific performance is required.
- **Backend:** Node.js with Express.js to build secure, RESTful, and scalable APIs.
- **Database:** PostgreSQL or MongoDB for reliable data storage and retrieval.
- **Authentication:** Firebase Authentication or JWT-based security for robust session verification.

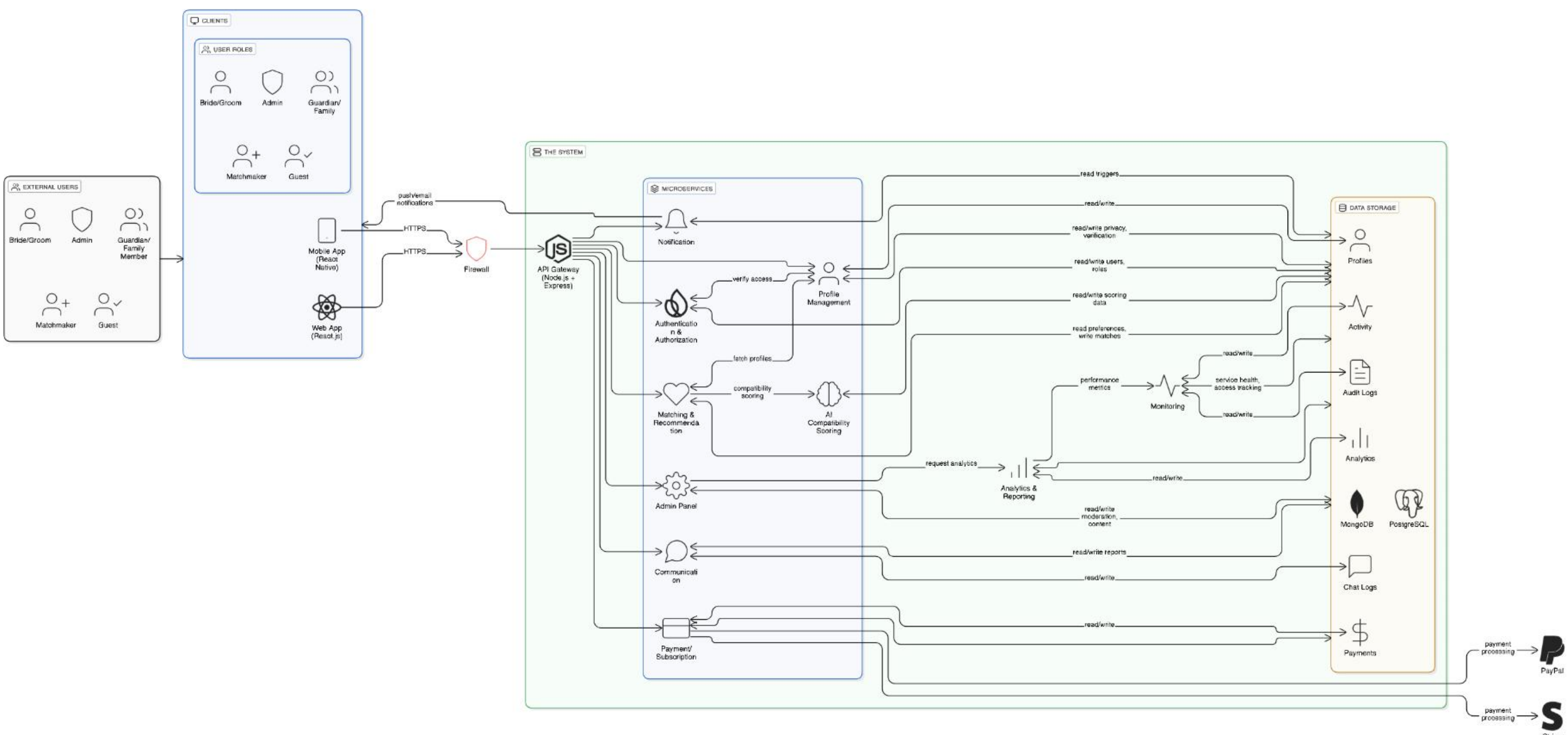
- **AI/ML:** TensorFlow Lite, Scikit-learn, and PyTorch Mobile for advanced recommendation algorithms.
- **Cloud & DevOps:** AWS/GCP for hosting, Docker for containerization, and GitHub Actions for CI/CD pipelines.

3.2 Key Architectural Components

The system is composed of several modular services accessible via an API Gateway:

1. **API Gateway:** Routes requests and handles load balancing.
2. **Profile Management:** Manages biodata, privacy settings, and user preferences.
3. **Verification Service:** Validates identity documents (CNIC, Passport) to assign "Verified" badges.
4. **Matchmaking Engine:** AI algorithms that calculate compatibility scores.
5. **Communication Service:** Handles real-time chat (WebSockets) and virtual video meetings.
6. **Notification Service:** Pushes alerts for interests, matches, and messages.
7. **Payment Module:** Processes subscriptions via secure gateways.

Software Architecture Diagram:



4. Core Modules & Functionality

4.1 User Profiles & Identity Verification

- **Secure Registration:** OTP-based signup.
- **Biodata Fields:** Detailed inputs for age, city, profession, religion, and family details.

- **Verification:** Users must upload Government-issued IDs (National ID, Driver's License, Passport). These are handled securely to award a verification badge, increasing profile visibility and trust.

4.2 Advanced Search & AI Matching

- **Filters:** Comprehensive filtering by age, education, community, and religion.
- **AI Compatibility:** Machine learning algorithms analyze user preferences to provide daily recommended matches and a "Compatibility Score."

4.3 Communication Suite

- **Interest System:** Users send "Interests" which must be "Accepted" before chatting.
- **Secure Chat:** In-app messaging prevents the premature sharing of personal phone numbers.
- **Virtual Meeting Rooms:** Built-in video/voice call features allow families to interact safely within the app before meeting in person.

4.4 Guardian Dashboard

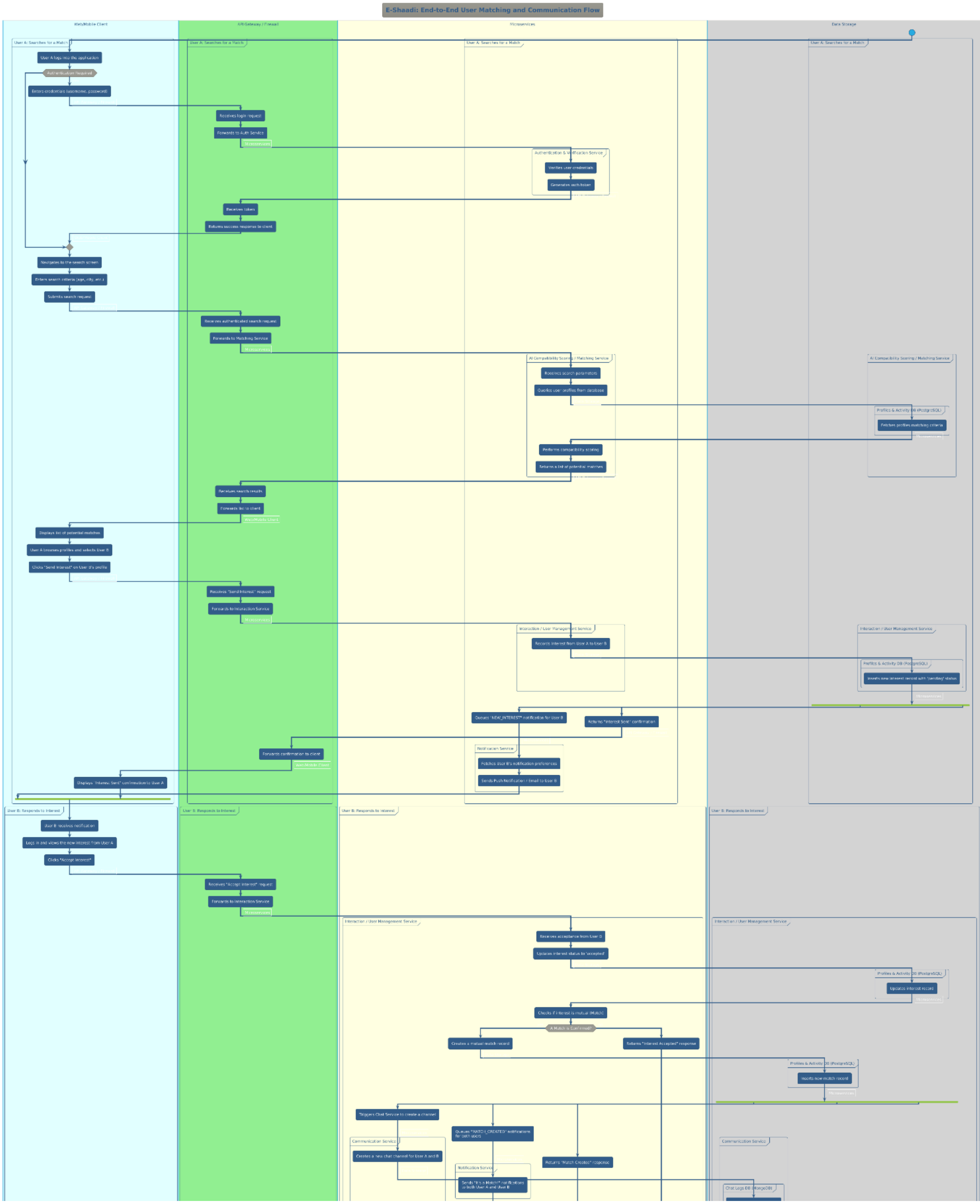
- **Family Management:** A unique feature allowing parents or guardians to manage profiles for their dependents.
- **Monitoring:** Guardians can track sent proposals, view proposal analytics, and schedule family meetings.

4.5 Community & Gamification

- **Events:** Users can connect through community-based groups.
- **Badges:** Gamification elements like "Most Viewed Profile," "Verified," and "Early Bird" encourage engagement.

5. System Design & Workflows

Activity Diagram:

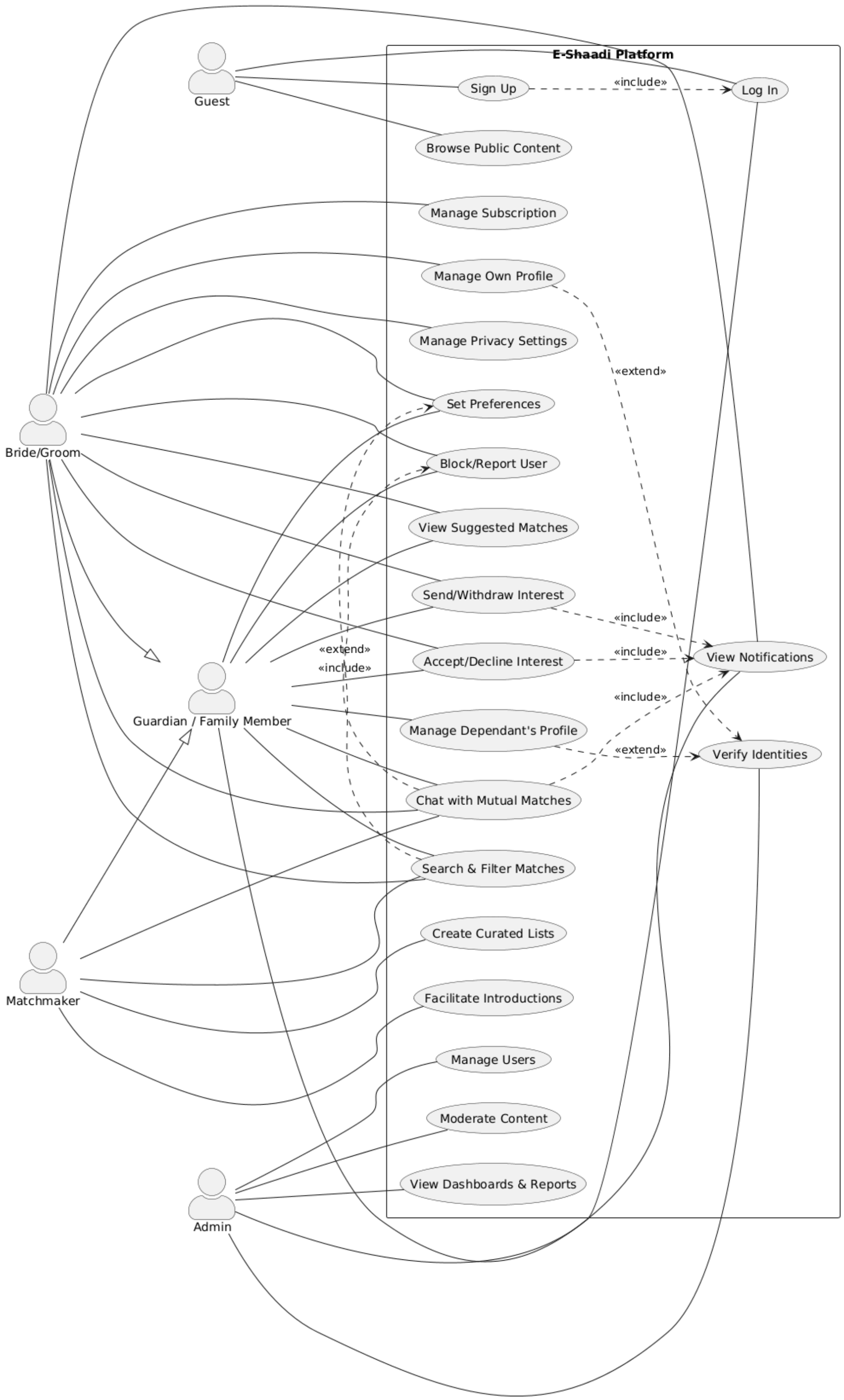


5.1 User Roles (Actors)

The system supports multiple distinct actors, each with specific permissions:

- **Guest:** Can sign up/log in and browse limited public content.
- **Bride/Groom:** The primary user who manages their profile, sets preferences, sends interests, and chats.
- **Guardian:** Manages a dependent's profile, filters matches, and facilitates introductions.
- **Matchmaker:** A professional or community role that creates curated lists and facilitates introductions.
- **Admin:** Manages users, moderates content, views analytics, and handles reports.

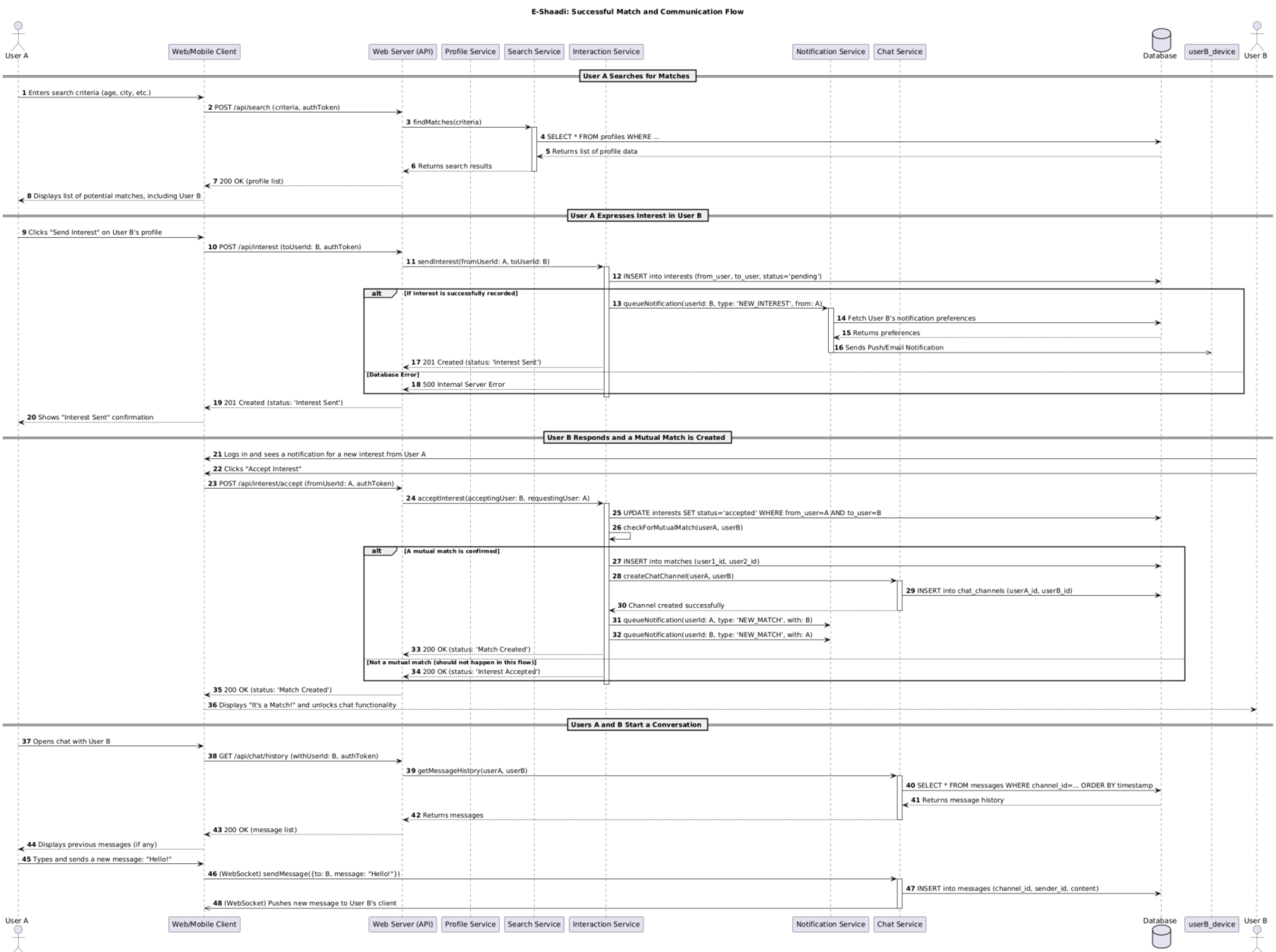
Use-Case Diagram:



5.2 Match & Communication Flow

1. **Search:** User A searches for matches based on criteria (age, city, etc.).
2. **Interest:** User A sends an "Interest" request to User B.
3. **Notification:** User B receives a notification (Push/Email).
4. **Action:** User B accepts the interest.
5. **Connection:** A mutual match is created, unlocking the chat functionality.
6. **Conversation:** Users interact via real-time WebSocket messaging.

Sequence Diagram:



6. Development Roadmap

The project is structured into three distinct phases to ensure quality and stability. The timeline below details the schedule.

Project Schedule

Phase	Core Activity	Duration	Task Details
Phase 1	**Planning & Design**	~60 Days	• System Analysis & Requirement Elicitation • Architecture Finalization
			• UI/UX Wireframing
Phase 2	**Development**	~90 Days	• **Backend:** API & Database Setup • **Frontend:** Mobile Screen Development • **DevOps:** CI/CD & Environment Setup
Phase 3	**Testing & Launch**	~45 Days	• QA & End-to-End Testing • App Store Submission & Deployment

- **Phase 1: Planning & Design**
  - System Analysis and Requirement Elicitation.
  - Architecture finalization (Mobile App -> API -> DB).
  - UI/UX wireframing and mockups.
- **Phase 2: Development**
  - **Backend:** API development, database schema design, and security implementation.

  - **Frontend:** Mobile screen development, component creation, and UI integration.
  - **DevOps:** CI/CD pipeline setup and environment configuration.
- **Phase 3: Testing & Deployment**
  - End-to-end testing, QA, and integration testing.
  - Final deployment to Google Play Store and Apple App Store.

Requirements Traceability Matrix (RTM): full matrix attached separately

Software Design and Analysis Project | Section 5J

Group: Muzammil Siddiqui, Rayyan Ali, Haseeb Asif, Azaan Hussain, Ahmed Ali

Module: All Modules Priority: All Priorities Status: All Status Search: Search requirements... Export to CSV

Total Requirements: **90** Displayed: **90** High Priority: **30** Coverage: **100%**

Req ID	Module	Requirement Description	Use Case(s)	Diagram Reference	Priority	Status	Test Scenario
REQ-001	Authentication	User must be able to register using email/phone with OTP verification	Sign Up, Log In	Main Use Case	High	Implemented	Verify OTP sent and account created successfully
REQ-002	Authentication	System must support role-based access control (Individual, Guardian, Admin, Matchmaker)	Sign Up, Manage Own Profile	Main Use Case	High	Implemented	Test different user roles have appropriate access
REQ-003	Authentication	User must verify email/phone before accessing full features	Verify Identities, Log In	Main Use Case, Verification	High	Implemented	Attempt login without verification
REQ-004	Profile Management	User must create profile with mandatory fields: name, age, gender, city, religion	Manage Own Profile	Main Use Case	High	Implemented	Submit profile with missing mandatory fields
REQ-005	Profile Management	User can add optional fields: education, profession, hobbies, future plans, family details	Manage Own Profile	Main Use Case	Medium	Implemented	Add and edit optional profile fields
REQ-006	Profile Management	User can upload multiple photos with privacy controls	Manage Own Profile, Manage Privacy Settings	Main Use Case	High	Implemented	Upload photos and set visibility controls
REQ-007	Profile Management	User can set privacy settings for photos and contact information	Manage Privacy Settings	Main Use Case	High	Implemented	Verify privacy settings restrict unauthorized access
REQ-008	Profile Management	Guardian can create and manage profiles for family members	Manage Dependant's Profile, Manage Multiple Profiles	Main Use Case, Guardian Dashboard	High	Implemented	Guardian creates profile for dependent
REQ-009	Search & Filter	User can perform basic search by age, city, religion, education	Search & Filter Matches	Main Use Case	High	Implemented	Search with multiple basic filters

7. Business Model

E-Shaadi employs a "Freemium" subscription model to monetize the platform while keeping basic features accessible.

Projected Revenue Distribution

Revenue Source	Share	Description
Premium Subscriptions	65%	Recurring revenue from Gold/Platinum members.
Profile Boosts	25%	One-time payments for temporary visibility spikes.
Community/Events	10%	Tickets for virtual or physical matchmaking events.

- **Revenue Streams:**
  - **Premium Subscriptions:** Unlocks unlimited swipes, "See Who Liked You," and advanced filters.
  - **Profile Boosts:** One-time payments to increase profile visibility.
- **Sample Pricing Tiers (Pitch Estimates):** ○ **1 Month:** ~499/mo
  - **6 Months:** ~2,250 total (Best Value)
  - **12 Months:** ~2,999 total (Save 50%)

8. Conclusion

E-Shaadi represents a significant step forward in the matrimonial technology space for the South Asian market. By combining the traditional values of family involvement and trust with the modern conveniences of AI matching and secure digital verification, the platform offers a robust solution to the age-old challenge of finding a life partner.